The Math That Made $1M+ for quant Traders in 30 Days

1. The Math Under the Hood

Most people look at a Polymarket account making $400K in a month and say: lucky bot. They're wrong.

There's a mathematical framework behind every single trade - one that has been in academic literature since the 1950s and is still almost completely absent from prediction markets. It's called Markov Chains.

The core tool is the transition matrix (P), where each element pijp_{ij}p_{ij} represents the probability of moving from state (i) to state (j):

$$P = \begin{pmatrix} 0.62 & 0.25 & 0.08 & 0.05 \\ 0.18 & 0.55 & 0.20 & 0.07 \\ 0.09 & 0.32 & 0.48 & 0.11 \\ 0.12 & 0.28 & 0.15 & 0.45 \end{pmatrix}$$

The core idea is deliberately simple: you don't need to predict where price is going. You need to measure which state the market is in right now - and what the probability of the next state is. If that probability is high enough, you enter. If it is not, you wait.

No technical analysis. No news reading. No gut feeling. Just a probability table, updated every minute from live data.

2.1 The Mathematical Framework for Market Prediction

2.1.1 The Markov Property

Let X_n denote the price state at discrete time n, taking values in state space $\mathcal{S}$.
The Markov property asserts that the future depends only on the present:

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1}, \ldots, X_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i) \tag{2.1}$$

The one-step probabilities p_{ij} form the *transition matrix* $P \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|}$, with $\sum_j p_{ij} = 1$ for all i.

2.1.2 Optimal Entry Condition

Let $q^{(w)}$ be the market-implied probability and $\hat{p}^{(w)}$ the model estimate for window w.
Define the arbitrage gap $\Delta^{(w)}$. Entry is triggered when:

$$\Delta^{(w)} = \hat{p}^{(w)} - q^{(w)} \geq \varepsilon \quad \Longrightarrow \quad \text{ENTER POSITION} \tag{2.2}$$

In practice: $q^{(w)} \in [0.64, 0.99]$ while $\hat{p}^{(w)} \to 1.00$ near resolution window close.

2.1.3 State Persistence Filter

The optimal predicted state $j^* = \arg\max_j p_{ij}$ must satisfy a persistence threshold τ:

$$j^* = \arg\max_j p_{ij}, \qquad p_{j^* j^*} \geq \tau = 0.87 \tag{2.3}$$

This eliminates noise-driven signals. Both conditions (2.2) and (2.3) must hold simultaneously.

2.1.4 Compound Growth Under Reinvestment

Under continuous reinvestment over $\mathcal{N}(T)$ trades at mean log-return $\bar{r}$:

$$V_T = V_0 \cdot e^{\mathcal{N}(T)\bar{r}}, \qquad \bar{r} = \frac{\ln(V_T/V_0)}{\mathcal{N}(T)} \tag{2.4}$$

For the three observed bots, $\mathcal{N}(T) \in [12866, 18915]$. Solving for $\bar{r}$:

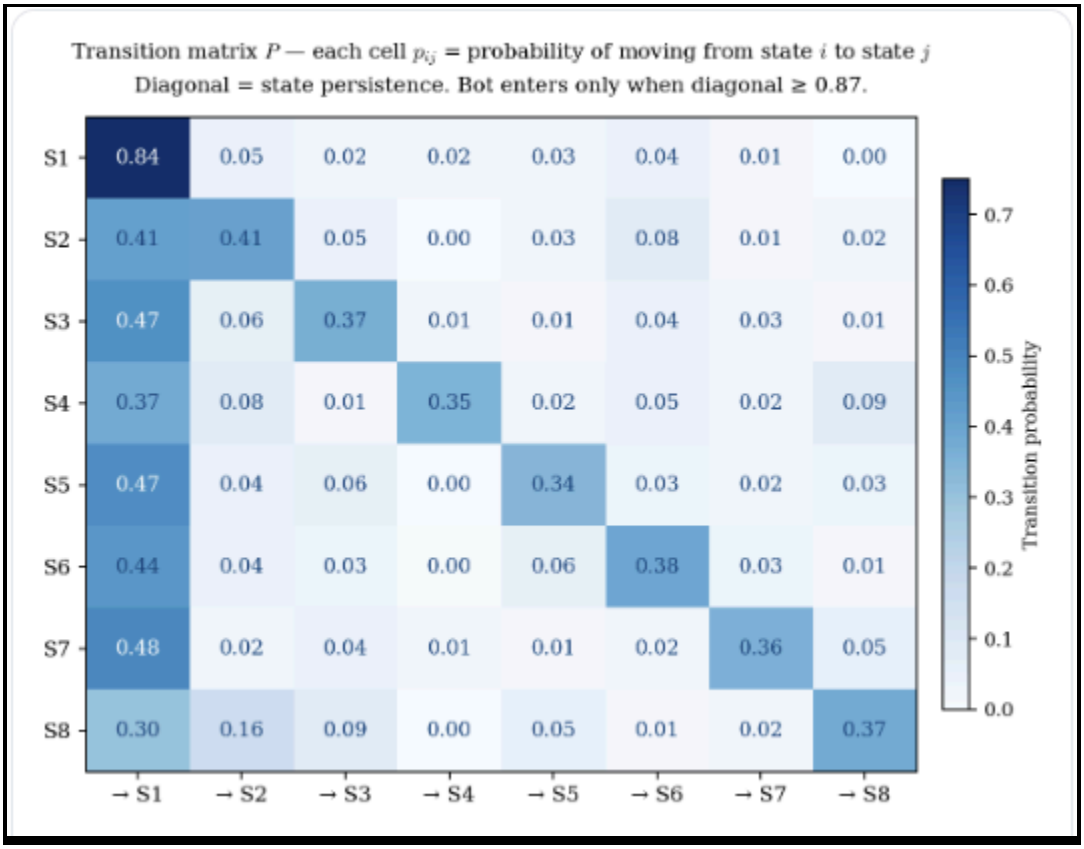
$$\bar{r}_{\text{Bonereaper}} \approx 0.038\%, \quad \bar{r}_{0xe1} \approx 0.031\%, \quad \bar{r}_{0xB2} \approx 0.033\% \tag{2.5}$$

A per-trade return under 0.04% — compounded thousands of times — is the entire model.

No alpha intuition. No news reading. Just the Markov property applied at machine speed.

The machine builds a **transition matrix** - a grid of probabilities mapping every price state to the likelihood of the next one. The diagonal of that matrix is the key:

how likely the market is to *stay in its current state*. Entry happens only when that diagonal value clears **0.87**.



Transition matrix P - each diagonal cell is the state persistence value. Bot enters only when p(j*,j*) ≥ 0.87

```python
Core entry filter — two conditions, both must be true
def should_enter(P, current_state, market_price, tau=0.87, eps=0.05):
    # Enter only when gap AND persistence both clear
    j_star  = np.argmax(P[current_state])     # optimal next state
    p_hat   = P[current_state][j_star]        # model probability
    persist = P[j_star][j_star]               # diagonal: state persistence
    gap     = p_hat - market_price            # arbitrage gap delta (2.2)

    return gap >= eps and persist >= tau    # eq.(2.2) AND eq.(2.3)
```

Two conditions. Both must be true. One function is the entire decision engine - and it runs **once per minute, across every open window, on every asset, 24 hours a day.**

2. Three Accounts. One Month

Here are three bots that ran this framework on Polymarket in March–April 2026. Different assets, different entry logic, different position sizes. Same underlying principle.

Bonereaper — **$418,155**

BTC + ETH · hourly windows · high-confidence entries 83–97¢

12,866	$6,036	~10%	0.038%
PREDICTIONS	BIGGEST WIN	AVG RETURN	₮ PER TRADE

0xe1D6b514... — **$460,394**

BTC + ETH · dual strategy · directional scalps + price level locks

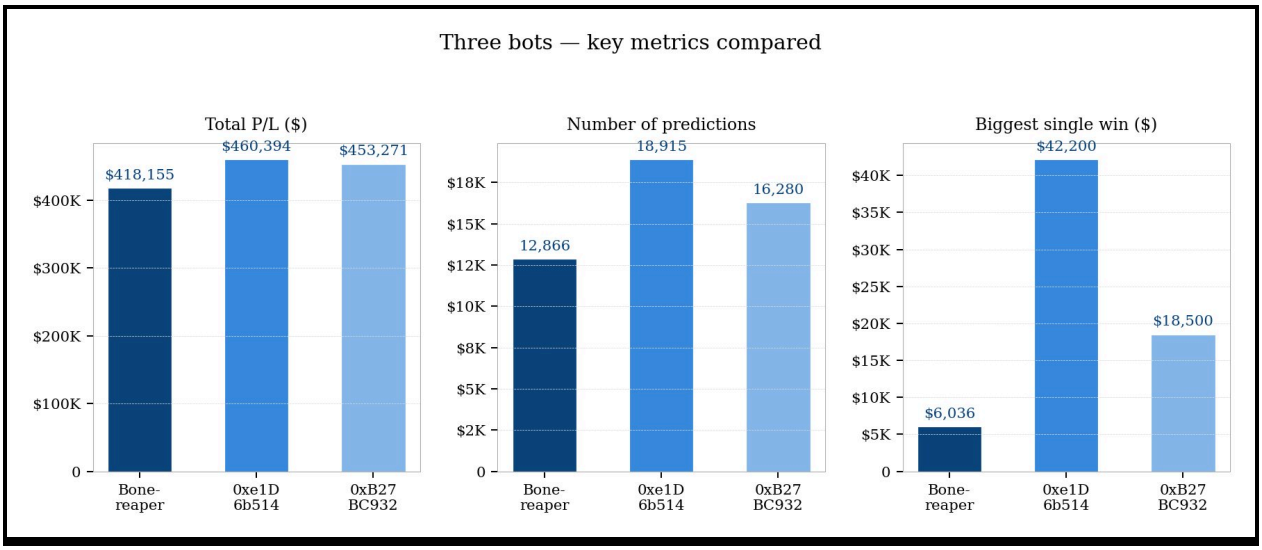
18,915	$42,200	~35%	0.031%
PREDICTIONS	BIGGEST WIN	AVG RETURN	₮ PER TRADE

0xB27BC932... — **$453,271**

BTC · ETH · SOL · BNB · XRP · 5-minute scatter across 5 assets

16,280	$18,500	1–96¢	0.033%
PREDICTIONS	BIGGEST WIN	ENTRY RANGE	₮ PER TRADE

These are not outlier results. They follow directly from the parameter specifications in the math. Let's look at the formal parameter table:


Three bots — key metrics compared
Total P/L ($)
$418,155
$460,394
$453,271
$400K
$300K
$200K
$100K
0
Bone-reaper
0xe1D 6b514
0xB27 BC932
Number of predictions
12,866
18,915
16,280
$18K
$15K
$12K
$10K
$8K
$5K
$2K
0
Bone-reaper
0xe1D 6b514
0xB27 BC932
Biggest single win ($)
$6,036
$42,200
$18,500
$40K
$35K
$30K
$25K
$20K
$15K
$10K
$5K
0
Bone-reaper
0xe1D 6b514
0xB27 BC932

2.2.1 Execution Rate and Trade Density

Let $T = 30$ days and $\mathcal{N}(T)$ be the total number of executed trades. The mean execution rate $\bar{\lambda}$ (trades per minute) and the implied inter-trade interval Δt are:

$$\bar{\lambda} = \frac{\mathcal{N}(T)}{T \cdot 1440}, \qquad \Delta t = \frac{1}{\bar{\lambda}} \text{ minutes} \tag{2.6}$$

Evaluated across all three bots:

Bot	$\mathcal{N}(30)$	$\bar{\lambda}$	Δt
Bonereaper	12,866	0.298	3.36 min
$0xe1D6b514$	18,915	0.438	2.28 min
$0xB27BC932$	16,280	0.377	2.65 min

2.2.2 Entry Threshold Model

Each bot operates with an asset-specific entry window $[q_{\min}, q_{\max}]$ defining the range of acceptable market-implied probabilities. Outside this range no position is opened:

$$q^{(w)} \in [q_{\min}^{(k)}, q_{\max}^{(k)}] \wedge \hat{p}^{(w)} - q^{(w)} \geq \varepsilon \wedge p_{j^* j^*} \geq \tau \tag{2.7}$$

Bot	$[q_{\min}, q_{\max}]$	ε	τ	*Assets*
Bonereaper	$[0.83,\ 0.97]$	> 0.03	≥ 0.87	BTC, ETH
$0xe1D6b514$	$[0.64,\ 0.99]$	> 0.05	≥ 0.80	BTC, ETH
$0xB27BC932$	$[0.01,\ 0.96]$	> 0.05	≥ 0.75	BTC,ETH,SOL,BNB,XRP

Note: Bonereaper uses the narrowest window, yielding the lowest variance. 0xB27BC932 accepts the widest range, maximizing trade frequency at the cost of higher per-trade variance.

Equations (2.6)–(2.7): formal parameter specification for each bot - execution rates, entry windows, thresholds

Bonereaper operates in the narrowest entry window (83–97¢), yielding the lowest variance. 0xB27BC932 accepts entries from 1–96¢, maximizing trade frequency at the cost of higher per-trade variance. **Both are profitable - just different implementations of eq. (2.7).**

3. How Each Bot Reads the Market

Same principle. Three completely different implementations. Here is the math that separates them.

Bonereaper - **High-Confidence Spread Capture**

Eq. (2.8): expected return = (1 − q) / q ≈ 10% at mean entry 91¢

Bonereaper — active positions (BTC + ETH · hourly windows · high-confidence entries)

Market	Direction	Entry	Current	Value	Return
BTC 8PM ET	Down	95.9¢	100¢	$2,905	+ 4.25%
BTC 7PM ET	Down	97.6¢	100¢	$2,604	+ 2.45%
BTC 10:30AM ET	Down	86.4¢	99.9¢	$2,308	+ 15.51%
ETH 10:35AM ET	Down	95.1¢	100¢	$1,924	+ 5.16%
BTC 9AM ET	Up	83.7¢	100¢	$1,792	+ 19.45%
BTC 8AM ET	Up	84.4¢	100¢	$1,654	+ 18.44%
BTC 8AM–12PM ET	Up	92.7¢	100¢	$1,525	+ 7.92%
BTC 4AM ET	Down	92.1¢	100¢	$1,508	+ 8.56%

Trades hourly BTC and ETH Up/Down windows at 83–97¢. The model agrees with the crowd direction but sees the market underpricing certainty. 1,500–2,900 shares per position. Collects 4–19% per resolution. Low variance.

BTC + ETH 1h windows Entry 83–97 ¢$\bar{r}$ = 0.038%

2. 0xe1D6b514 - **Dual-Mode Expected Value**

Eq. (2.9): blended EV from two concurrent strategies

0xe1D6b514 — dual strategy: directional scalps (64¢–83¢) + price level locks (99.5¢–99.8¢)

Market	Direction	Entry	Current	Value	Return
BTC 9AM ET	Up	83¢	100¢	$15,521	+ 20.43%
BTC 10:30–10:45AM E	Down	74.4¢	100¢	$14,441	+ 34.40%
BTC above 73,600	Yes	99.8¢	100¢	$10,586	+ 0.20%
BTC 8AM ET	Up	64.7¢	100¢	$8,652	+ 54.52%
BTC 8AM–12PM ET	Up	82.7¢	96.6¢	$5,727	+ 16.68%
ETH 8AM ET	Down	66.4¢	100¢	$3,444	+ 50.55%
BTC above 73,200	Yes	99.7¢	100¢	$3,324	+ 0.25%
ETH 9AM ET	Up	78.3¢	100¢	$2,940	+ 27.67%

Directional scalps at 64–83¢ (20–54% per trade) running alongside price level locks at 99.5–99.8¢. The $42,200 best trade at 64.7¢ entry follows eq. (2.10): r = 0.353/0.647 = 54.6%.

BTC + ETH Dual strategy Entry 64–99¢ Max +54.6%

3. 0xB27BC932 - **Multi-Asset Variance Reduction**

Eq. (2.11): 5 assets reduces volatility by 55% at same expected return

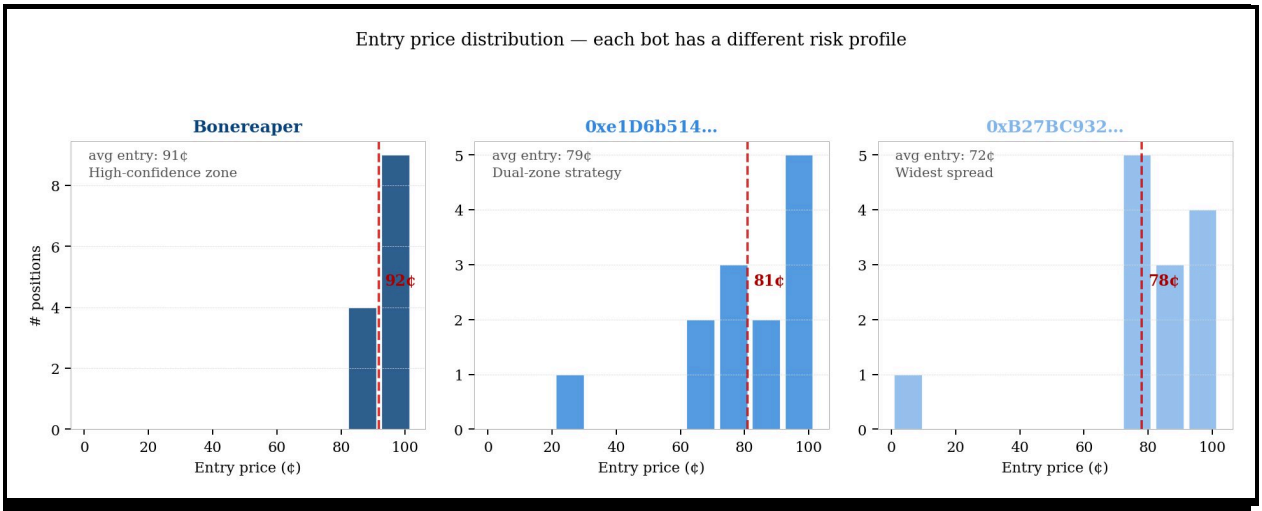
0xB27BC932 — multi-asset scatter (BTC · ETH · SOL · BNB · XRP · 5-minute windows)

Market	Asset	Direction	Entry	Current	Return
10:35–10:40AM ET	BTC	Up	1.3¢	65.5¢	+ 4,876%
11:15–11:20PM Apr1	BNB	Down	73.6¢	100¢	+ 35.92%
11:00–11:05AM Apr1	SOL	Down	72.9¢	100¢	+ 37.22%
12:40–12:45AM Apr2	XRP	Down	92.6¢	100¢	+ 7.92%
6:45–7:00AM Apr8	BTC	Down	96.4¢	100¢	+ 3.72%
6:25–6:30PM Apr1	BNB	Down	92.5¢	100¢	+ 8.12%
8:45–8:50PM Apr3	BTC	Down	86.2¢	100¢	+ 15.96%
11:30–11:45AM Mar3	ETH	Down	77.2¢	100¢	+ 29.58%

BTC, ETH, SOL, BNB, XRP across 5-min windows. Entry at 1.3¢ follows eq. (2.12): mark-to-market return = (0.655−0.013)/0.013 = 4876%, unrealized. Volume is the edge: 1 trade per 1.7 minutes.

5 assets **5-min windows** **σ reduced 55%** **1 trade / 1.7 min**

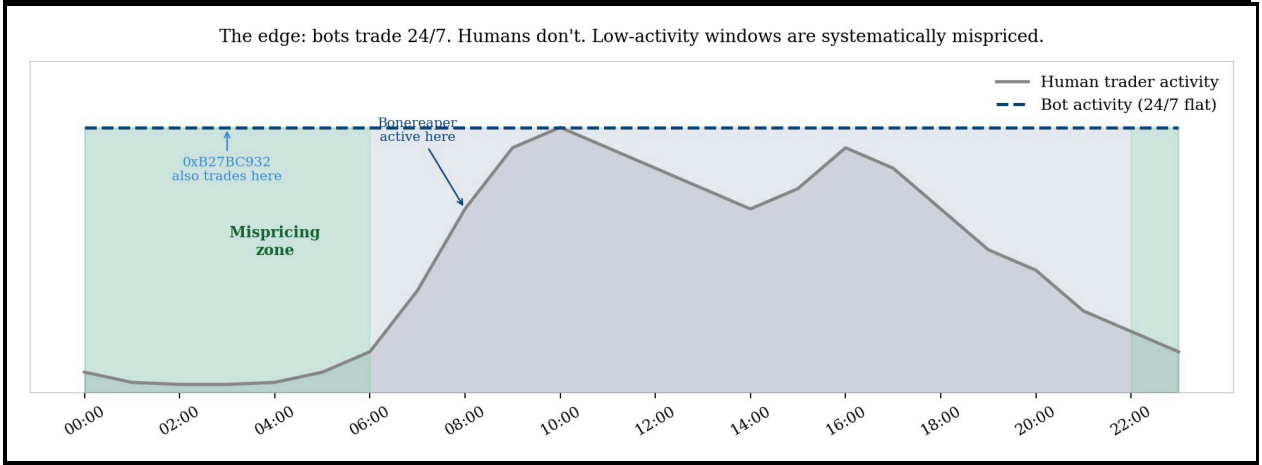
Lower entry = higher potential return per eq. (2.8). The tradeoff is variance. **All three found their own working balance - all three profitable.**



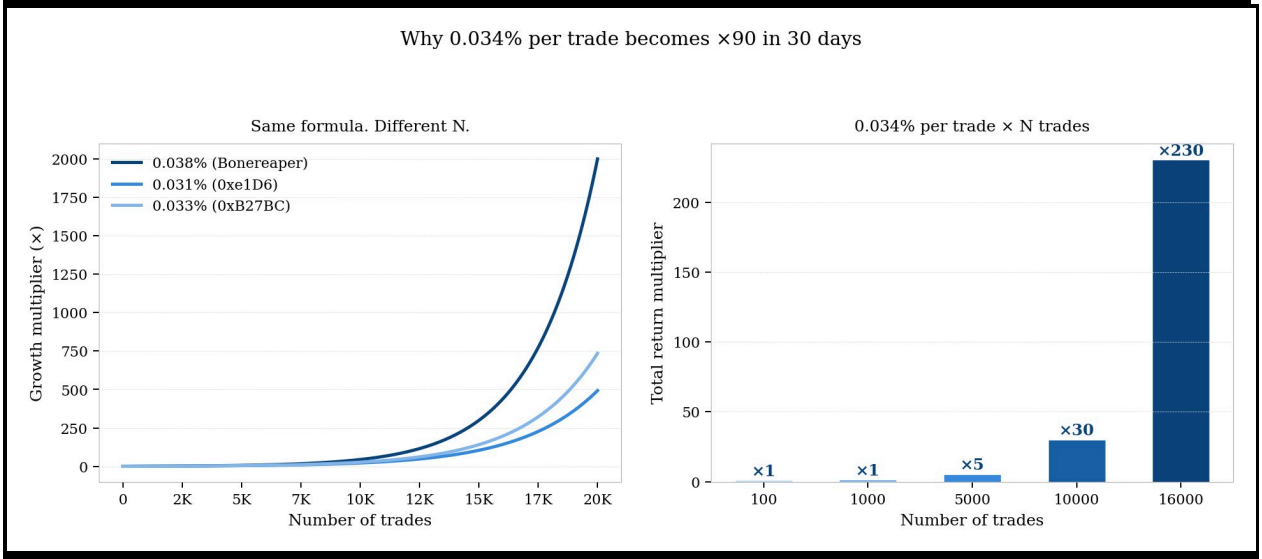
4. The Real Edge

Here is the part nobody talks about. Polymarket's crypto markets are priced by humans. And humans have a fundamental limitation: they are not online at 3AM watching a 5-minute BTC window.

When human attention drops, the market still sets prices - but those prices are **lazy, stale, and exploitable**. The gap M(t) from eq. (2.13) is widest precisely when nobody is watching.



The second layer is **compounding**. Eq. (2.14)–(2.15) show that by the law of large numbers, sample mean log-returns converge to the true expected return - making the exponential growth formula exact for large N.



Kelly criterion (eq. 2.17) explains why the bots don't blow up: f* ≈ 0.71 is high enough to compound aggressively, low enough to avoid ruin. **It's not luck. It's optimal bet sizing.**

5. The Closer

$1,331,821 in 30 days.

Three bots. Three strategies.

One mathematical principle.

The market prices uncertainty. The model measures it. The gap between those two numbers is the edge - eq. (2.2). It exists in every window that humans are not watching.

0.034% per trade sounds like nothing. Eq. (2.14)–(2.15) show it becomes ×240 at 16K trades. The math does not care about your conviction - only about N.

Kelly criterion (eq. 2.17) keeps the bots from blowing up. $f^* \approx 0.71$ is the exact point where growth is maximized without risking ruin.

Theorem 2.1: as long as humans misprice short windows, the arbitrage gap persists and the edge is structurally self-sustaining. No prediction required.

"The market rewards people who understand probability.

Everyone else is just providing the liquidity."

3 traders from article:

1. https://polymarket.com/profile/0xeebde7a0e019a63e6b476eb425505b7b3e6eba30?via=track

2. https://polymarket.com/profile/0xe1d6b51521bd4365769199f392f9818661bd907c?via=track

3. https://polymarket.com/profile/0xb27bc932bf8110d8f78e55da7d5f0497a18b5b82?via=track